

Semester III

Core-V: Data Base Management System

Course Objectives:

- To understand the database concepts for efficient storage and retrieval of data.
- To learn about database design and transaction processing

Learning Outcomes:

Upon completion of this course, students will be able to:

- Build data models using entity relationship concepts
- Design databases by systematically applying the normalization process
- Create relational database tables and perform various operations using SQL
- Learn issues relating to database transactions and approaches to deal with them

Unit I:

Introduction to Database and Database Users, Database System Concepts and Architecture: data Models, schema, and instances, Conceptual Modeling and Database Design, Entity Relationship (ER) Model: Entity Types, Entity Sets, Attributes, Keys, Relationship Types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, ER Naming Conventions. Enhanced Entity-Relationship (EER) Model.

Unit II:

Relational data Model and SQL: Relational Model Concepts, Basic SQLs, SQL Data Definition and Data types, Constraints in SQL, Retrieval Queries in SQL, INSERT, DELETE, UPDATE Statements in SQL, Relational Algebra and Relational Calculus: Unary Relational Operations: SELECT and PROJECT, Binary Relation: JOIN and DIVISION.

Unit III:

Database Design Theory and Normalization: Functional Dependencies, Normal Forms based on Primary Keys, Second and third Normal Forms, Boyce-Codd Normal Form, Multivalued Dependency and Fourth Normal Form, Join Dependencies and Fifth Normal Form.

Unit IV:

Transaction Processing Concepts: Transaction and System Concepts, Properties of Transactions, Recoverability, Serializability, Concurrency Control Techniques, Locking techniques for Concurrency Control, Concurrency Control based on Time-Stamp Ordering.

Text Books:

- ✓ *Fundamentals of Database Systems, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson Education*
- ✓ *Database Management Systems, Rajiv Chopra, S. Chand Pubs.*

Reference Book:

- ✓ *An Introduction to Database System, Date C. J. - Pearson Education, New Delhi*

Core-V Lab: Data Base Management System

Create and use the following database schema to answer the given queries.

EMPLOYEE Schema

Field	Type	NULL	KEY	DEFAULT
Eno	Char(3)	NO	PRI	NIL
Ename	Varchar(50)	NO		NIL
Job_type	Varchar(50)	NO		NIL
Manager	Char(3)	Yes	FK	NIL
Hire_date	Date	NO		NIL
Dno	Integer	YES	FK	NIL
Commission	Decimal(10,2)	YES		NIL
Salary	Decimal(7,2)	NO		NIL

DEPARTMENT Schema

Field	Type	NULL	KEY	DEFAULT
Dno	Integer	No	PRI	NULL
Dname	Varchar(50)	Yes		NULL
Location	Varchar(50)	Yes		New Delhi

List of Queries:

1. Display Employee Name, Job, Hire Date, Employee Number; for each employee with the Employee Number appearing first.
2. Display unique Jobs from the Employee Table.
3. Display the Employee Name concatenated by a Job separated by a comma.
4. Display all the data from the Employee Table. Separate each Column by a comma and name the said column as THE_OUTPUT.
5. Display the Employee Name and Salary of all the employees earning more than \$2850.
6. Display Employee Name and Department Number for the Employee No= 7900.
7. Display Employee Name and Salary for all employees whose salary is not in the range of \$1500 and \$2850.
8. Display Employee Name and Department No. of all the employees in Dept 10 and Dept 30 in the alphabetical order by name.
9. Display Name and Hire Date of every Employee who was hired in 1981.
10. Display Name and Job of all employees who don't have a current Manager.
11. Display the Name, Salary and Commission for all the employees who earn commission.
12. Sort the data in descending order of Salary and Commission.
13. Display Name of all the employees where the third letter of their name is 'A'.
14. Display Name of all employees either have two 'R's or have two 'A's in their name and are either in Dept No = 30 or their Managers Employee No = 7788.
15. Display Name, Salary and Commission for all employees whose Commission Amount is 14 greater than their Salary increased by 5%.
16. Display the Current Date.
17. Display Name, Hire Date and Salary Review Date which is the 1st Monday after six months of employment.

18. Display Name and calculate the number of months between today and the date each employee was hired.
19. Display the following for each employee <E-Name> earns < Salary> monthly but wants <3*Current Salary>. Label the Column as Dream Salary.
20. Display Name with the 1st letter capitalized and all other letter lower case and length of their name of all the employees whose name starts with 'J', 'A' and 'M'.
21. Display Name, Hire Date and Day of the week on which the employee started.
22. Display Name, Department Name and Department No for all the employees.
23. Display Unique Listing of all Jobs that are in Department # 30.
24. Display Name, Department Name of all employees who have an 'A' in their name.
25. Display Name, Job, Department No. and Department Name for all the employees working at the Dallas location.
26. Display Name and Employee no. Along with their Manger's Name and the Manager's employee no; along with the Employees Name who do not have a Manager.
27. Display Name, Department No. And Salary of any employee whose department no. and salary matches both the department no. And the salary of any employee who earns a commission.
28. Display Name and Salaries represented by asterisks, where each asterisk (*) signifies \$100.
29. Display the Highest, Lowest, Sum and Average Salaries of all the employees.
30. Display the number of employees performing the same Job type functions.
31. Display the no. of managers without listing their names.
32. Display the Department Name, Location Name, No. of Employees and the average salary for all employees in that department.
33. Display Name and Hire Date for all employees in the same dept. as Blake.
34. Display the Employee No. And Name for all employees who earn more than the average salary.
35. Display Employee Number and Name for all employees who work in a department with any employee whose name contains a 'T'.
36. Display the names and salaries of all employees who report to King.
37. Display the department no, name and job for all employees in the Sales department.